一千六百年前的药方

屠呦呦和青蒿素的故事

A 1,600-Year-Old Recipe

Tu Youyou and the story of artemisinin

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Footnotes mark names, festival terms and words above HSK 6 (HSK 3.0 lists: github.com/ivankra/hsk30, MIT).

1 “青蒿一握[†]，以水二升渍[‡]，绞取汁[§]，尽服之[¶]。”这是一千六百多年前一本医书里的一句话，一共只有十五个字。在很长的时间里，几乎没有人特别注意过它。直到二十世纪七十年代，一位中国女科学家在一堆发黄的旧书里又读到了它。从这十五个字出发，她和她的同事们找到了一种新药。据估计，这种药后来救了几百万人的生命。

二〇一五年十月

2 2015年10月5日晚上，许晴[*]坐在奶奶家的沙发上看手机。那是国庆假期，她从北京回到了淮河[#]边的老家。她是北京一所大学药学系二年级的学生。这个假期她本来打算好好复习，可是心里一直很乱。同宿舍的女孩已经决定转到金融专业，还劝她也一起转。

† 青蒿一握 *qīnghāo yī wò* take a handful of qinghao

‡ 以水二升渍 *yǐ shuǐ èr shēng zì* soak it in two sheng of water

§ 绞取汁 *jiǎo qǔ zhī* wring out the juice

¶ 尽服之 *jìn fú zhī* drink all of it

* 许晴 *Xǔ Qíng* Xu Qing

淮河 *Huái Hé* the Huai River

"Take a handful of qinghao, soak it in two sheng of water, wring out the juice and drink all of it." These are words from a medical book written more than 1,600 years ago: just fifteen characters in all. For a very long time, hardly anyone paid them any special attention. Then, in the 1970s, a Chinese woman scientist came across them again in a pile of yellowed old books. Starting from those fifteen characters, she and her colleagues found a new medicine. It is estimated that this medicine has since saved millions of lives. 1

October 2015

On the evening of 5 October 2015, Xu Qing was sitting on the sofa at her grandmother's house, looking at her phone. It was the National Day holiday, and she had come home from Beijing to her family's village by the Huai River. She was a second-year pharmacy student at a university in Beijing. She had meant to study hard over the holiday, but her mind had been in a muddle the whole time. The girl who shared her dorm room had already decided to switch to finance, and kept urging her to switch too. 2

2 “学药有什么意思？天天在实验室里跟老鼠在一起，做一百次实验，九十九次都失败。”室友这样说的时候，许晴[1]没有回答，因为她自己也说不清，学药到底有什么意思。

3 奶奶在厨房里洗碗，电视开着，正在放一部老电影。突然，手机上跳出一条新闻：中国科学家屠呦呦[2]获得了2015年诺贝尔[3]生理学或医学奖。许晴一下子坐直了。这个名字她在课本上见过，就在讲青蒿素[4]的那一页。新闻说，屠呦呦今年八十四岁，她和同事们发现的青蒿素，是治疗疟疾[5]的重要药物；她也是第一位获得诺贝尔科学奖的中国女科学家。“奶奶，快来看！中国科学家得诺贝尔奖[6]了！”

4 奶奶擦着手走过来，戴上眼镜，看了半天，只认出了两个字：“疟疾？就是打摆子[7]吧？”

1 许晴 *Xǔ Qíng* Xu Qing
2 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
3 诺贝尔 *Nuòbèi'ěr* Nobel
4 青蒿素 *qīnghāosù* artemisinin
5 疟疾 *nüèjí* malaria
6 诺贝尔奖 *Nuòbèi'ěrjiǎng* the Nobel Prize
7 打摆子 *dǎbǎizi* to have malaria (dialect)

"What's the point of studying pharmacy? Every day in the lab with the mice, and ninety-nine experiments out of a hundred fail." When her roommate said this, Xu Qing didn't answer, because she couldn't have said herself what the point of pharmacy really was. 2

Grandma was washing the dishes in the kitchen. The television was on, showing an old film. Suddenly a news alert popped up on her phone: the Chinese scientist Tu Youyou had won the 2015 Nobel Prize in Physiology or Medicine. Xu Qing sat bolt upright. She had seen the name in her textbook, on the very page about artemisinin. The news said that Tu Youyou was eighty-four, that the artemisinin she and her colleagues had discovered was an important drug against malaria, and that she was the first Chinese woman scientist to win a Nobel Prize in science. "Grandma, come and look! A Chinese scientist has won the Nobel Prize!" 3

Grandma came over wiping her hands, put on her glasses and peered at the screen for a long time, but recognised only two characters. "Malaria? That's the shaking sickness, isn't it?" 4

4 “打摆子[†]？”“我们这儿的人都这么叫。”奶奶在她旁边坐了下来，声音低了下去，“那个病啊，我小时候见得多了。”电视上的电影还在放，可是谁也没有再看它。

疟疾[‡]

5 奶奶说，她小时候，村子四周都是水田和小河。一到夏天，蚊子[§]多得像一团一团的烟。晚上大家坐在院子里乘凉，手里的扇子一刻也不敢停。可是不管怎么扇，蚊子照样咬人。“那时候，谁知道蚊子会传病啊？大家都说，是天太热，水不干净。”

6 奶奶十二岁那年夏天，她的弟弟病了。一开始只是冷，冷得全身发抖[¶]，盖了三床被子，还是不停地发抖，牙碰得直响。

† 打摆子 *dǎbǎizi* to have malaria (dialect)
‡ 疟疾 *nüèjí* malaria
§ 蚊子 *wénzi* mosquito
¶ 发抖 *fādǒu* to shiver

"The shaking sickness?" "That's what everyone round here calls it." Grandma sat down beside her, and her voice dropped. "That illness... I saw plenty of it when I was little." The film on the television played on, but neither of them was watching it any more. 4

Malaria

Grandma said that when she was a child, the village was surrounded by paddy fields and streams. As soon as summer came, there were so many mosquitoes that they hung in the air like clouds of smoke. In the evenings everyone sat in the courtyard to enjoy the cool air, never daring to stop their fans for a moment. But however hard they fanned, the mosquitoes bit them all the same. "Back then, who knew that mosquitoes spread disease? Everyone said it was the heat and the dirty water." 5

The summer Grandma was twelve, her little brother fell ill. At first he was only cold, so cold that his whole body shook. Even under three quilts he couldn't stop shivering, and his teeth chattered. 6

6 过了一两个小时，他又热得像火烧一样，满头大汗†，嘴里说着谁也听不懂的话。热退了以后，人便像一张纸一样躺在床上，一点儿力气也没有。隔了一天，同样的事又来一遍。“一冷一热，一冷一热，就像钟摆一样，摆过来，又摆过去，所以我们叫它打摆子‡。”奶奶说。

7 那时候村里没有医院，只有一个赤脚医生§。他背着药箱走了十几里路赶来，给弟弟吃了几片白色的小药片，苦得弟弟直哭。弟弟后来好了，可是村里有的孩子没有这么幸运。“那年秋天，后面那家的小女儿，就没了。”奶奶看着电视，很久没有说话。许晴¶也不知道该说什么。她从来没听奶奶讲过这些。

† 满头大汗 *mǎntóu dàhàn* dripping with sweat

‡ 打摆子 *dǎbǎizi* to have malaria (dialect)

§ 赤脚医生 *chìjiǎo yīshēng* “barefoot doctor”, village medic

¶ 许晴 *Xǔ Qíng* Xu Qing

An hour or two later he was burning up as if on fire, his head dripping with sweat, muttering words no one could understand. When the fever broke, he lay on the bed as limp as a sheet of paper, without any strength at all. A day later, the whole thing would happen again. "Cold, then hot, cold, then hot, like a pendulum swinging this way and that. That's why we call it the shaking sickness," said Grandma. 6

In those days there was no hospital in the village, only a "barefoot doctor". He walked several miles with his medicine kit on his back and gave her brother a few little white pills, so bitter that the boy cried and cried. Her brother got better in the end, but some of the village children were not so lucky. "That autumn, the little girl from the family behind us... we lost her." Grandma looked at the television and said nothing for a long time. Xu Qing didn't know what to say either. She had never heard Grandma talk about any of this. 7

8 许晴[1]在课上学过：疟疾[2]是由疟原虫[3]引起的。蚊子[4]咬了病人，再去咬别人，疟原虫就从一个人身上到了另一个人身上。这种病跟人类的历史一样古老。据说，三千多年前的甲骨文[5]里，就已经有“疟[6]”字了。十七世纪，欧洲人[7]知道了南美洲[8]有一种树的皮能治这种病。后来，人们从这种树皮里提取出了奎宁[9]，又做出了氯喹[10]。奶奶说的那几片白色小药片，很可能就是氯喹。可是到了二十世纪六十年代，很多地方的疟原虫对氯喹产生了抗药性[11]，这种药便渐渐失去了作用。全世界的科学家都在寻找新的药。

1 许晴 *Xǔ Qíng* Xu Qing
2 疟疾 *nüèjí* malaria
3 疟原虫 *nüèyuánchóng* malaria parasite
4 蚊子 *wénzi* mosquito
5 甲骨文 *jiǎgǔwén* oracle bone script
6 疟 *nüè* malaria (the character)
7 欧洲人 *Ōuzhōurén* Europeans
8 南美洲 *Nán Měizhōu* South America
9 奎宁 *kuíníng* quinine
10 氯喹 *lǜkuí* chloroquine
11 抗药性 *kàngyàoxìng* drug resistance

Xu Qing had learned in class that malaria is caused by a parasite. When a 8
mosquito bites a sick person and then bites someone else, the parasite passes from one body to the next. The disease is as old as human history. The character for malaria is said to appear already in oracle bone inscriptions more than three thousand years old. In the seventeenth century, Europeans learned that the bark of a South American tree could treat the disease. Later quinine was extracted from that bark, and then chloroquine was made. The little white pills Grandma remembered were very likely chloroquine. But by the 1960s the parasites in many places had become resistant to chloroquine, and the drug gradually stopped working. Scientists all over the world were searching for a new medicine.

五二三[†]

9 那时候，越南[‡]正在进行一场战争。在又湿又热的山林里，疟疾[§]比枪和子弹更可怕。据说，因为疟疾而不能战斗的士兵，比在战斗中受伤的还多。越南向中国请求帮助。1967年5月23日，一个秘密的全国性研究项目在北京开始了。为了保密，它没有正式的名字，人们便用开会的日子来叫它："五二三"。在后来的十几年里，全国几十个单位、几百名科研人员先后参加了这个项目。有的人合成新的化学药物，有的人去中草药里寻找。可是两年过去了，进展并不大。

10 1969年1月，北京的中医研究院[¶]也接到了任务。研究院让一位三十九岁的女研究人员担任课题组组长。她就是屠呦呦[*]。那时候，她在研究院已经工作了十几年，可是在研究院外面，几乎没有人知道她的名字。

† 五二三 *Wǔ'èrsān* Project 523
‡ 越南 *Yuènán* Vietnam
§ 疟疾 *nüèjí* malaria
¶ 中医研究院 *Zhōngyī Yánjiūyuàn* Academy of Traditional Chinese Medicine
* 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)

Project 523

At that time, a war was being fought in Vietnam. In the hot, humid forests, 9
malaria was more frightening than guns and bullets. It is said that more soldiers were put out of action by malaria than were wounded in battle. Vietnam asked China for help. On 23 May 1967, a secret nationwide research project began in Beijing. To keep it secret it had no official name, so people simply called it by the date of the meeting: "523". Over the following decade and more, several hundred researchers from dozens of institutions across the country took part in the project. Some made new synthetic drugs; others searched among Chinese medicinal herbs. But two years went by with little progress.

In January 1969, the Academy of Traditional Chinese Medicine in Beijing 10
was given the task as well. The Academy appointed a thirty-nine-year-old woman researcher to lead the research group. She was Tu Youyou. By then she had worked at the Academy for more than ten years, but outside it hardly anyone knew her name.

10 据说，研究院选她，是因为她既学过现代药学，又认真学过中医，这在当时并不多见。

11 屠呦呦[1]1930年生于浙江[2]宁波[3]。她的名字是父亲起的，来自《诗经[4]》里的一句诗：“呦呦鹿鸣[5]，食野之苹[6]。”意思是：鹿[7]呦呦[8]地叫着，在田野里吃草。有意思的是，这首诗的下一段写的是“呦呦鹿鸣，食野之蒿[9]”——蒿[10]，正是青蒿[11]的蒿。十六岁那年，她得了肺结核，不得不停学两年。据说，正是这场病，让她对医药产生了兴趣。1951年，她考上了北京大学[12]医学院的药学系。毕业以后，她到中医研究院[13]工作，后来又花了两年半的时间，系统地学习中医。

1 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
2 浙江 *Zhèjiāng* Zhejiang province
3 宁波 *Níngbō* Ningbo, a city in Zhejiang
4 诗经 *Shījīng* the Book of Songs
5 呦呦鹿鸣 *yōuyōu lù míng* “yoyo”, cry the deer
6 食野之苹 *shí yě zhī píng* eating the wild herbs
7 鹿 *lù* deer
8 呦呦 *yōuyōu* (a deer’s call)
9 食野之蒿 *shí yě zhī hāo* eating the wild hao
10 蒿 *hāo* wormwood (Artemisia)
11 青蒿 *qīnghāo* qinghao, sweet wormwood
12 北京大学 *Běijīng Dàxué* Peking University
13 中医研究院 *Zhōngyī Yánjiūyuàn* Academy of Traditional Chinese Medicine

It is said that the Academy chose her because she had studied modern pharmacy and had also seriously studied Chinese medicine, which was not common at the time. 10

Tu Youyou was born in 1930 in Ningbo, Zhejiang. Her father chose her name from a line in the Book of Songs: "Yoyo, cry the deer, eating the wild herbs." It means: the deer call "yoyo" as they graze in the fields. Curiously, the next verse of the poem reads "Yoyo, cry the deer, eating the wild hao" — and hao is the very hao of qinghao. At sixteen she caught tuberculosis and had to stop school for two years. It is said that this illness was what made her interested in medicine. In 1951 she won a place in the pharmacy department of Peking University's medical school. After graduating she went to work at the Academy of Traditional Chinese Medicine, and later spent two and a half years studying Chinese medicine systematically. 11

12 接到任务的时候，她的丈夫正被派到外地，两个女儿都还很小。为了工作，她把大女儿送进了托儿所，把小女儿送回了宁波[†]老家。据说，几年以后小女儿回到北京，已经不认识自己的妈妈了。“当妈的，心里该多难受啊。”奶奶轻轻地说。

两千个药方[‡]

13 屠呦呦[§]的小组决定从古书开始。中国人跟疟疾[¶]斗争了几千年，过去的医生一定留下了不少经验。他们翻遍了历代的医书，走访了很多老中医，还读了各地寄来的信和民间药方。据说，只用了大约三个月，他们就收集了两千多个药方。屠呦呦从中选出了六百四十个，编成了一本小册子。每一个药方，都写清楚了用什么药、怎么用、来自哪本书。

† 宁波 *Níngbō* Ningbo, a city in Zhejiang
‡ 药方 *yàofāng* prescription, recipe
§ 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
¶ 疟疾 *nüèjí* malaria

When she was given the task, her husband had been sent away to another province, and their two daughters were still very small. For the sake of the work, she put her elder daughter into a nursery and sent the younger one back to the family home in Ningbo. It is said that when the younger girl came back to Beijing a few years later, she no longer recognised her own mother. "As a mother, how hard that must have been," Grandma said softly. 12

Two Thousand Recipes

Tu Youyou's group decided to start with the old books. The Chinese had been fighting malaria for thousands of years; the doctors of the past must have left a good deal of experience behind. They searched through the medical books of every dynasty, went to see many old doctors of Chinese medicine, and read the letters and folk remedies sent in from all over the country. It is said that in only about three months they had collected more than two thousand recipes. From these Tu Youyou chose six hundred and forty and compiled them into a booklet. For every recipe it was written down clearly which herbs it used, how they were used and which book it came from. 13

14 接下来是一次又一次的实验。他们把草药做成提取物[†]，给身上有疟原虫[‡]的老鼠吃，然后检查老鼠血液里的疟原虫有没有减少。据记载，前后两年多，他们试了一百多种中药、三百八十多个提取物。结果大多让人失望。有的药第一次好像有点儿效果，再做一次，效果就没有了。

15 青蒿[§]也在名单上。青蒿是一种很普通的野草，田边、路旁、山坡上到处都有。它的叶子细细的，有一股特别的味道。第一次实验的时候，青蒿让老鼠身上的疟原虫减少了百分之六十八，大家都很兴奋。可是再试几次，结果却只有百分之十二到百分之四十。效果时好时坏，谁也说不清为什么。因而，不少人觉得，青蒿大概也没什么希望了。

16 那几年，项目里的很多单位都在经历失败。

† 提取物 *tíqǔwù* extract

‡ 疟原虫 *nüèyuánchóng* malaria parasite

§ 青蒿 *qīnghāo* qinghao, sweet wormwood

What followed was experiment after experiment. They turned the herbs into 14
extracts and fed them to mice infected with the parasite, then checked whether the number of parasites in the mice's blood had gone down. According to the records, over a little more than two years they tried more than a hundred Chinese herbs and over three hundred and eighty extracts. Most of the results were disappointing. Some herbs seemed to have a little effect the first time, but when the experiment was repeated, the effect was gone.

Qinghao was on the list too. Qinghao is a very common wild plant that 15
grows everywhere, beside fields, along roads and on hillsides. Its leaves are fine and thin, and it has a distinctive smell. In the first experiment, qinghao cut the parasites in the mice by sixty-eight per cent, and everyone was excited. But when they tried a few more times, the results were only twelve to forty per cent. The effect was sometimes good and sometimes poor, and nobody could say why. As a result, many people felt that qinghao probably held little promise either.

In those years, many of the institutions in the project were going through 16
failure after failure.

16 有的人被调走了，有的人开始怀疑：几千年前的老办法，真的能对付今天的疟原虫[1]吗？屠呦呦[2]照样每天早早来到实验室，很晚才离开。她想，青蒿[3]的效果时好时坏，问题也许不在青蒿本身，而在提取的方法上。“那她怎么办呢？”奶奶问。“她回去读古书了。”许晴[4]说。

一句古文

17 屠呦呦又一次打开了那些古书。这一次，她的目光停在了东晋[5]医生葛洪[6]写的《肘后备急方[7]》上。葛洪是一位道士，也是一位医生，大约生活于公元三、四世纪。“肘后[8]”的意思是，书小得可以放在袖子里，随时带在身边；“备急”就是为急病准备的。

1 疟原虫 *nüèyuánchóng* malaria parasite
2 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
3 青蒿 *qīnghāo* qinghao, sweet wormwood
4 许晴 *Xǔ Qíng* Xu Qing
5 东晋 *Dōng Jìn* Eastern Jin dynasty (317–420)
6 葛洪 *Gě Hóng* Ge Hong, 4th-century physician
7 肘后备急方 *Zhǒuhòu Bèijí Fāng* Emergency Recipes to Keep at Hand
8 肘后 *zhǒuhòu* “behind the elbow”

Some people were transferred elsewhere, and some began to doubt: could 16
methods thousands of years old really deal with today's malaria parasites? Tu Youyou carried on as before, arriving at the lab early every day and leaving late. Since the effect of qinghao came and went, she thought, perhaps the problem lay not in the plant itself but in the way it was extracted. "So what did she do?" asked Grandma. "She went back to the old books," said Xu Qing.

One Line of Classical Chinese

Once again Tu Youyou opened the old books. This time her eyes came to rest 17
on Emergency Recipes to Keep at Hand, written by the Eastern Jin physician Ge Hong. Ge Hong was a Daoist and also a physician, who lived around the third and fourth centuries AD. "Behind the elbow" means the book was small enough to slip into a sleeve and carry about at all times; "ready for emergencies" means it was meant for sudden illness.

17 这本书大约写于一千六百多年前，是写给老百姓看的。书里的药大多简单、便宜，在田边路旁就能找到。在讲治疟疾[1]的那一部分里，有这样一句话：“青蒿一握[2]，以水二升渍[3]，绞取汁[4]，尽服之[5]。”意思是：拿一把青蒿[6]，用两升水泡着，再把它的汁[7]挤出来，全部喝下去。

18 这句话，屠呦呦[8]以前也读过。可是这一次，她注意到了一个以前没有想过的问题。过去的人用中药，一般都要放在锅里煮，可是葛洪[9]写的不是“煮”，而是“渍[10]”和“绞[11]”——用冷水泡，再把汁挤出来。从头到尾[12]，青蒿一直没有被加热。她想到，问题也许就出在温度上。他们以前提取青蒿，用的是水煮，或者用酒精[13]加热的办法。

1 疟疾 *nüèjí* malaria
2 青蒿一握 *qīnghāo yī wò* take a handful of qinghao
3 以水二升渍 *yǐ shuǐ èr shēng zì* soak it in two sheng of water
4 绞取汁 *jiǎo qǔ zhī* wring out the juice
5 尽服之 *jìn fú zhī* drink all of it
6 青蒿 *qīnghāo* qinghao, sweet wormwood
7 汁 *zhī* juice
8 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
9 葛洪 *Gě Hóng* Ge Hong, 4th-century physician
10 渍 *zì* to soak
11 绞 *jiǎo* to wring
12 从头到尾 *cóngtóu dàowěi* from beginning to end
13 酒精 *jiǔjīng* alcohol, ethanol

The book was written about 1,600 years ago, for ordinary people. Most of the 17
remedies in it were simple and cheap, things you could find beside a field or a road. In the part on treating malaria there is this line: "Take a handful of qinghao, soak it in two sheng of water, wring out the juice and drink all of it." That is: take a handful of qinghao, steep it in two sheng of water, squeeze out the juice, and drink it all down.

Tu Youyou had read this line before. But this time she noticed something 18
she had never thought about. In the past, Chinese herbs were usually boiled in a pot. But Ge Hong did not write "boil": he wrote "soak" and "wring" — steep the plant in cold water, then squeeze out the juice. From beginning to end, the qinghao was never heated. It occurred to her that the problem might lie in the temperature. When they had extracted qinghao before, they had boiled it in water or heated it with alcohol.

18 也许，青蒿[†]里真正有用的成分怕热，一遇到高温便被破坏了。凭着这句一千六百年前的古文，她决定换一种办法：改用乙醚[‡]。乙醚的沸点[§]只有三十五度左右，比水和酒精[¶]都低得多。用它在较低的温度下提取，以便让那种可能怕热的成分保存下来。

19 听到这里，奶奶指了指窗外："青蒿？我们后山上多的是。早知道这样，那年给我弟弟挤一碗喝，不就好了？""可不能这么想。"许晴[*]连忙说，"古书只是给了她一个线索。真正的药，是后来一步一步做出来的，要经过很多实验和检查。""自己找点儿草来喝，不但治不好病，还可能让病变得更重。生了病，还是得去看医生。"奶奶笑了："好，好，你是学药的，听你的。"

† 青蒿 *qīnghāo* qinghao, sweet wormwood
‡ 乙醚 *yǐmí* ether
§ 沸点 *fèidiǎn* boiling point
¶ 酒精 *jiǔjīng* alcohol, ethanol
* 许晴 *Xǔ Qíng* Xu Qing

Perhaps the truly useful substance in qinghao could not stand heat, and was 18
destroyed as soon as it met high temperatures. On the strength of this 1,600-year-old line, she decided to try a different method: to use ether instead. Ether boils at only about thirty-five degrees, much lower than water or alcohol. With it they could extract at a lower temperature, so that the substance which might be sensitive to heat would survive.

At this, Grandma pointed out of the window. "Qinghao? There's plenty of 19
it on the hill behind us. If only we'd known, that year we could have squeezed a bowl of it for my brother, couldn't we?" "You mustn't think like that," Xu Qing said quickly. "The old book only gave her a clue. The real medicine was made later, step by step, and it had to go through a great many experiments and tests." "Picking some plants and drinking them yourself won't cure anything, and it might make the illness worse. If you're ill, you still have to see a doctor." Grandma laughed. "All right, all right. You're the one studying pharmacy. I'll listen to you."

第一百九十一号

20 换了办法以后，成功并没有马上到来。样品[1]一个一个地做，一个一个地写上号码，又一个一个地失败。实验室的条件很差，没有专门的设备。据说，他们就买来了几口普通的大水缸[2]，用来泡青蒿[3]。乙醚[4]有毒，味道很大，还很容易着火，因而每一步都得特别小心。屋子里的空气又很不好，时间一长，很多人都头疼、流鼻血。屠呦呦[5]后来还得了肝病。

21 1971年10月，第一百九十一号样品送到了实验台上。这是一份用乙醚从青蒿里提取出来的样品。结果出来的时候，大家都不敢相信自己的眼睛：吃了这种提取物[6]的老鼠，血液里的疟原虫[7]几乎全部不见了。效果达到了百分之百。后来在猴子身上做实验，结果照样是百分之百。

1 样品 *yàngpǐn* sample
2 水缸 *shuǐgāng* water vat
3 青蒿 *qīnghāo* qinghao, sweet wormwood
4 乙醚 *yǐmí* ether
5 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
6 提取物 *tíqǔwù* extract
7 疟原虫 *nüèyuánchóng* malaria parasite

Number 191

Changing the method did not bring success right away. The samples were made one by one, numbered one by one, and failed one by one. Conditions in the lab were poor, and there was no proper equipment. It is said that they simply bought a few ordinary big water vats and used them to soak the qinghao. Ether is poisonous, has a strong smell and catches fire very easily, so every step had to be taken with special care. The air in the rooms was bad too, and after a while many of them suffered headaches and nosebleeds. Tu Youyou later developed liver trouble as well. 20

In October 1971, sample number 191 arrived on the lab bench. It was an extract made from qinghao with ether. When the result came out, nobody could believe their eyes: in the mice that had taken the extract, the parasites in the blood had almost completely disappeared. The effect was one hundred per cent. Later, when they ran the experiment on monkeys, the result was again one hundred per cent. 21

22 “第一百九十一号”——这个数字后来常常被人提起。它告诉人们的事很简单：在它前面，还有很多没有成功的样品[1]。许晴[2]讲到这里，忽然想起了室友的话：做一百次实验，九十九次都失败。如果在第一百九十号的时候，大家都放弃了呢？1972年3月，屠呦呦[3]在南京[4]召开的五二三[5]项目会议上报告了这个结果，引起了很大的注意。

自己先试

23 可是，新的问题马上来了：这种提取物[6]用在人身上，安全吗？在一些动物实验里，出现了让人担心的情况，有人怀疑它可能有毒。要弄清楚，还需要做很多检查，这要花很长的时间。可是疟疾[7]有自己的季节。一旦错过了这一年的夏天和秋天，在病人身上的试验便要再等一年。

1 样品 *yàngpǐn* sample
2 许晴 *Xǔ Qíng* Xu Qing
3 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
4 南京 *Nánjīng* Nanjing
5 五二三 *Wǔ'èrsān* Project 523
6 提取物 *tíqǔwù* extract
7 疟疾 *nüèjí* malaria

"Number 191" — the figure was often mentioned afterwards. What it tells us is simple: before it there were a great many samples that did not succeed. When she reached this point, Xu Qing suddenly remembered her roommate's words: ninety-nine experiments out of a hundred fail. What if everyone had given up at number 190? In March 1972, Tu Youyou reported the result at a Project 523 meeting held in Nanjing, and it drew a great deal of attention. 22

Testing It on Themselves

But a new problem came up at once: was the extract safe to use on people? Some of the animal experiments had shown worrying signs, and some people suspected it might be toxic. Finding out would take many more tests, and that would take a long time. But malaria has its own season. If they missed that year's summer and autumn, trials on patients would have to wait another year. 23

24 于是，屠呦呦[1]向领导提出：她愿意自己先试。据报道，她的理由很简单：她是组长，应该第一个试。1972年7月，她和两位同事住进了北京的一家医院，每天按时吃青蒿[2]提取物[3]，由医生给他们做检查。一个星期过去了，他们的身体没有出现明显的问题。后来，又有几位同事试了，吃的量更大，结果照样安全。

25 奶奶吃惊地问："拿自己试药？她就不怕吗？"许晴[4]想了想，说："书上没写她怕不怕。我只知道，那年秋天，她带着药去了海南[5]。"海南是当时疟疾[6]很严重的地方。在那里，他们用这种提取物治疗了二十多名病人。病人的烧很快就退了，血里的疟原虫[7]也不见了。

1 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
2 青蒿 *qīnghāo* qinghao, sweet wormwood
3 提取物 *tíqǔwù* extract
4 许晴 *Xǔ Qíng* Xu Qing
5 海南 *Hǎinán* Hainan Island
6 疟疾 *nüèjí* malaria
7 疟原虫 *nüèyuánchóng* malaria parasite

So Tu Youyou told her superiors that she was willing to try it on herself first. According to reports, her reason was simple: she was the group leader, so she should be the first to try it. In July 1972, she and two colleagues checked into a hospital in Beijing. They took the qinghao extract at set times every day, and doctors examined them. A week passed, and none of them showed any obvious problems. Later several more colleagues tried it, taking larger amounts, and it again proved safe. 24

Grandma was astonished. "Trying the medicine on herself? Wasn't she afraid?" Xu Qing thought about it. "The books don't say whether she was afraid. All I know is that in the autumn of that year she took the medicine to Hainan." Hainan was a place where malaria was very serious at the time. There they treated more than twenty patients with the extract. The patients' fevers came down quickly, and the parasites vanished from their blood. 25

26 1972年11月，屠呦呦[1]的小组从青蒿[2]里得到了一种无色的结晶[3]——这才是真正起作用的东西。后来，人们叫它“青蒿素[4]”。再后来，北京和上海的科学家一起弄清了它的化学结构，成果发表于1977年。同一时期，云南[5]、山东[6]等地的研究人员也做了很多重要的工作。研究青蒿素的，不是一个人，而是许多单位的许多人。屠呦呦获奖以后也多次说过，这个奖不只属于她一个人，也属于整个团队。

屋子里的灯

27 奶奶听完，半天没说话。过了一会儿，她问：“那她得这个奖，等了多少年？”“从发现青蒿素到今天，四十多年了。”许晴[7]说。

1 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
2 青蒿 *qīnghāo* qinghao, sweet wormwood
3 结晶 *jiéjīng* crystal
4 青蒿素 *qīnghāosù* artemisinin
5 云南 *Yúnnán* Yunnan province
6 山东 *Shāndōng* Shandong province
7 许晴 *Xǔ Qíng* Xu Qing

In November 1972, Tu Youyou's group obtained a colourless crystal from 26
qinghao — the substance that really did the work. It was later named "artemisinin". Later still, scientists in Beijing and Shanghai worked out its chemical structure together, and the results were published in 1977. In the same period, researchers in Yunnan, Shandong and elsewhere also did a great deal of important work. Artemisinin was studied not by one person but by many people from many institutions. After winning the prize, Tu Youyou said many times that it belonged not just to her but to the whole team.

The Light in the Room

When Xu Qing had finished, Grandma said nothing for a long while. After 27
a time she asked, "So how many years did she wait for this prize?" "It's been more than forty years since artemisinin was discovered," said Xu Qing.

她又告诉奶奶，屠呦呦[1]没有博士学位，没有出国留过学，也不是院士[2]，所以有人叫她“三无科学家”。奶奶笑了：“没有那些，人家照样救了那么多人。”

那年12月，屠呦呦在斯德哥尔摩[3]用中文做了一次演讲，题目是《青蒿素[4]——中医药给世界的一份礼物》。那时，以[5]青蒿素为基础的联合治疗，已经是世界卫生组织[6]建议的主要治疗方法。可是，疟疾[7]并没有消失。直到今天，全世界每年仍然有两亿多人得疟疾，几十万人死于这种病，其中大部分是非洲[8]五岁以下的孩子。在东南亚[9]等地，一些疟原虫[10]已经开始对青蒿素产生抗药性[11]。一千六百年前的那句话，为人们打开了一扇门；可是门后面，还有更多的门。

1 屠呦呦 *Tú Yōuyōu* Tu Youyou (born 1930)
2 院士 *yuànshì* academician
3 斯德哥尔摩 *Sīdégē'ěrmó* Stockholm
4 青蒿素 *qīnghāosù* artemisinin
5 以 *yǐ* taking (X as Y)
6 世界卫生组织 *Shìjiè Wèishēng Zǔzhī* World Health Organization
7 疟疾 *nüèjí* malaria
8 非洲 *Fēizhōu* Africa
9 东南亚 *Dōngnányà* Southeast Asia
10 疟原虫 *nüèyuánchóng* malaria parasite
11 抗药性 *kàngyàoxìng* drug resistance

She also told Grandma that Tu Youyou had no doctorate, had never studied abroad and was not an academician, so some people called her the "scientist with three withouts". Grandma smiled. "Without any of that, she saved all those people just the same." 27

That December, Tu Youyou gave a lecture in Chinese in Stockholm, entitled "Artemisinin: A Gift from Traditional Chinese Medicine to the World". By then, combination treatments based on artemisinin were the main treatment recommended by the World Health Organization. But malaria has not disappeared. Even today, more than two hundred million people around the world catch malaria every year, and several hundred thousand die of it, most of them African children under five. In Southeast Asia and elsewhere, some parasites have already begun to resist artemisinin. That line from 1,600 years ago opened a door; but behind the door there are more doors still. 28

29 那天夜里，奶奶早早睡了。许晴[1]一个人坐在小屋里，拿出了那本很久没翻的药理学课本。她在第一页上慢慢地写下了那十五个字：青蒿一握[2]，以水二升渍[3]，绞取汁[4]，尽服之[5]。然后，她给室友发了一条短信："我不转了。"屋子里很静，只有桌上的台灯亮着。她忽然想，四十多年前，在北京那间满是乙醚[6]味道的屋子里，大概也常常有一个窗口亮着灯，一直亮到很晚很晚。

30 六年以后，2021年6月，世界卫生组织[7]宣布：中国已经消除了疟疾[8]。那天，许晴正在一家药物研究所的实验室里做实验。她放下手里的东西，给奶奶打了个电话。奶奶在电话那头听了很久，最后只说了一句："好啊。以后的孩子，都不知道什么叫打摆子[9]了。"

1 许晴 *Xǔ Qíng* Xu Qing
2 青蒿一握 *qīnghāo yī wò* take a handful of qinghao
3 以水二升渍 *yǐ shuǐ èr shēng zì* soak it in two sheng of water
4 绞取汁 *jiǎo qǔ zhī* wring out the juice
5 尽服之 *jìn fú zhī* drink all of it
6 乙醚 *yǐmí* ether
7 世界卫生组织 *Shìjiè Wèishēng Zǔzhī* World Health Organization
8 疟疾 *nüèjí* malaria
9 打摆子 *dǎbǎizi* to have malaria (dialect)

That night Grandma went to bed early. Xu Qing sat alone in her little room 29
and took out the pharmacology textbook she hadn't opened in a long time. On the first page she slowly wrote out those fifteen characters: take a handful of qinghao, soak it in two sheng of water, wring out the juice and drink all of it. Then she sent her roommate a text: "I'm not switching." The room was very quiet; only the desk lamp was on. It suddenly struck her that more than forty years ago, in that room in Beijing full of the smell of ether, there had probably often been a window lit up like this, until very late at night.

Six years later, in June 2021, the World Health Organization announced that 30
China had eliminated malaria. That day Xu Qing was in the laboratory of a drug research institute, running experiments. She put down what she was holding and phoned her grandmother. Grandma listened for a long time at the other end of the line, and in the end said only one thing: "Good. Children from now on won't even know what the shaking sickness is."